

```
ClearAll[g, u, ε, k, N];
```

 **ClearAll:** Symbol N is Protected.

```
eqs = {
  ε g + 8 k (N + 8) g^2 + 2 N k u^2 == 0,
  ε u + 16 k (N + 2) g u + 16 k u^2 == 0
};
```

```
sol = Solve[eqs, {g, u}, Assumptions → k < 0]
```

Out[120]=

$$\left\{ \{g \rightarrow 0, u \rightarrow 0\}, \left\{ g \rightarrow -\frac{\varepsilon}{16 k (4 + N)}, u \rightarrow -\frac{\varepsilon}{8 k (4 + N)} \right\}, \right. \\ \left. \left\{ g \rightarrow -\frac{\varepsilon}{8 k (8 + N)}, u \rightarrow 0 \right\}, \left\{ g \rightarrow -\frac{N \varepsilon}{16 k (8 + N^2)}, u \rightarrow \frac{(-4 + N) \varepsilon}{8 k (8 + N^2)} \right\} \right\}$$

In[122]:=

```
ClearAll[g, u, ε, N];
```

 **ClearAll:** Symbol N is Protected.

In[123]:=

```
βg = ε g + 8 (N + 8) g^2 + 2 N u^2;
βu = ε u + 16 (N + 2) g u + 16 u^2;
```

In[125]:=

```
(* Jacobian / stability matrix M_ij = ∂_{x_j} β_i *)
M = D[{βg, βu}, {{g, u}}] // Simplify;
```

In[126]:=

```
fplList = {
  {g → 0, u → 0},
  {g → -ε / (16 (4 + N)), u → -ε / (8 (4 + N))},
  {g → -ε / (8 (8 + N)), u → 0},
  {g → -(N ε) / (16 (8 + N^2)), u → (-4 + N) ε / (8 (8 + N^2))}
};
```

In[127]:=

```
Table[
  With[{Mstar = Simplify[M /. fp, Assumptions →  $\epsilon > 0$  &&  $N > 0$ ]},
    {
      "Fixed point" → fp,
      "Eigenvalues" → Eigenvalues[Mstar] // Simplify
    }
  ],
  {fp, fpList}
] // Column
```

Out[127]=

```
{Fixed point → {g → 0, u → 0}, Eigenvalues → { $\epsilon$ ,  $\epsilon$ }}
{Fixed point → {g →  $-\frac{\epsilon}{16(4+N)}$ , u →  $-\frac{\epsilon}{8(4+N)}$ }, Eigenvalues → { $\frac{(-2+N)\epsilon}{4+N}$ ,  $-\epsilon$ }}
{Fixed point → {g →  $-\frac{\epsilon}{8(8+N)}$ , u → 0}, Eigenvalues → { $-\epsilon$ ,  $-\frac{(-4+N)\epsilon}{8+N}$ }}
{Fixed point → {g →  $-\frac{N\epsilon}{16(8+N^2)}$ , u →  $\frac{(-4+N)\epsilon}{8(8+N^2)}$ }, Eigenvalues → { $-\epsilon$ ,  $\frac{(8-6N+N^2)\epsilon}{8+N^2}$ }}
```

In[128]:=

```
ClearAll[g, u,  $\epsilon$ , N];
```

 **ClearAll**: Symbol N is Protected.

In[129]:=

```
 $\beta g = \epsilon g + 8(N + 8)g^2 + 2Nu^2;$ 
 $\beta u = \epsilon u + 16(N + 2)gu + 16u^2;$ 
M = D[{ $\beta g$ ,  $\beta u$ }, {{g, u}}];
```

In[132]:=

```
fpAssoc = <|
  "Gaussian" → {g → 0, u → 0},
  "Isotropic" → {g →  $-\epsilon/(16(4+N))$ , u →  $-\epsilon/(8(4+N))$ },
  "Ising" → {g →  $-\epsilon/(8(8+N))$ , u → 0},
  "Cubic" → {g →  $-(N\epsilon)/(16(8+N^2))$ ,
    u →  $(-4+N)\epsilon/(8(8+N^2))$ }
|>;
```

In[133]:=

```

evList = Table[
  Module[{fp = fpAssoc[label]},
    {label,
      Simplify[Eigenvalues[M /. fp] /.  $\epsilon \rightarrow 1$ , Assumptions  $\rightarrow N > 0$ ]}
  ],
  {label, Keys[fpAssoc]}
];

```

In[134]:=

```

makePlot[label_String, { $\lambda_1$ _,  $\lambda_2$ _}] :=
  Plot[
    Evaluate[{ $\lambda_1$ ,  $\lambda_2$ }, {N, 0.5, 8},
    PlotLegends  $\rightarrow$  {" $\lambda_1/\epsilon$ ", " $\lambda_2/\epsilon$ "},
    PlotLabel  $\rightarrow$  Row[{label, " fixed point"}],
    PlotRange  $\rightarrow$  {-2, 2},
    AxesLabel  $\rightarrow$  {"N", None},
    Epilog  $\rightarrow$  {Red, Dashed, Line[{0.5, 0}, {8, 0}]}
  ];

```

In[135]:=

```

plots = (makePlot @@@ evList);
GraphicsGrid[Partition[plots, 2]]

```

Out[136]=

